

Ethical Risk Assessment of Predictive AI Models in Healthcare Logistics



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Declaration

I declare that this dissertation is my own original work and has been composed solely by myself. All sources of information used in this research have been duly referenced and acknowledged. This work has not been submitted, in whole or in part, for any other degree or qualification at this or any other institution. Any contributions from external tools, datasets, or third-party libraries have been clearly indicated.

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Research Topic: Ethical Risk Assessment of Predictive AI Models in Healthcare Logistics

Abstract

Predictive artificial intelligence planning models are increasingly being used in hospital capacity planning, intensive care unit resource allocation and hospital supply chain decisions. Though these models have enabled improvements to hospital operations, they have raised ethical concerns over bias, transparency and equity of performance across patients and hospitals. This Applied Research Project will implement a workflow to assess these ethical risks using publicly reported hospital capacity data.

A one hundred thousand row sample of the US Department of Health and Human Services COVID-19 Reported Patient Impact and Hospital Capacity by Facility dataset was pre-processed and feature engineered. Features were filtered to include only statistically informative features using a chi-square test of independence. Seven base models were trained using Logistic Regression, Decision Tree, Random Forest, XGBoost, Gradient Increasing, Extra Trees, and LightGBM. A stacking ensemble, consisting of the five tree-based learners, is built with the help of Logistic Regression as the meta-learner. All models are evaluated on the training dataset using stratified five-fold cross-validation and on the held-out test dataset comprising thirty percent of the overall dataset. The best-performing tree-based model is examined using SHAP explainability.

The chi-square test performed on the following features: hospital state, number of beds, number of ICU beds, number of used ICU beds and ICU beds occupancy rate, indicated that these features were statistically considerably correlated with the binary target of ICU capacity risk at the standard five percent importance level. On the holdout test set, LightGBM, Stacking

Ensemble and Gradient Increasing models achieved the highest F1 score of 0.9992. The SHAP analysis also identified the occupancy rate as the most important predictor, followed by the number of ICU beds, ICU beds occupied and bed occupancy rate, although individual state-level statistics were less informative, collectively providing useful information.

The results inform four sub-questions on predictive performance, hospital group differences, the interpretability of non-transparent models, as well as the risks of using biased model recommendations as an ethical basis for the redistribution of scarce healthcare resources. The study introduce the workflow as a reusable methodology that healthcare data teams can apply to their own data ecosystem to operationalize governance principles from WHO ethics guidance and the European Union AI Act.the study also discuss limitations and opportunities for future work including multi-region datasets, live monitoring during deployment, and formal governance frameworks.

Keywords: predictive AI, healthcare logistics, ethical risk assessment, machine learning, stacking ensemble, chi-square test, SHAP, fairness, transparency, ICU capacity.

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List of Abbreviations

AI	Artificial Intelligence
AUC	Area Under the Curve
CSV	Comma-Separated Values
CV	Cross-Validation
EHR	Electronic Health Record
EU	European Union
F1	Harmonic mean of precision and recall
GBM	Gradient Boosting Machine
HHS	United States Department of Health and Human Services
ICU	Intensive Care Unit
KDD	Knowledge Discovery and Data Mining
KNN	K-Nearest Neighbours
LightGBM	Light Gradient Boosting Machine
LIME	Local Interpretable Model-Agnostic Explanations
LR	Logistic Regression
ML	Machine Learning
RF	Random Forest
ROC	Receiver Operating Characteristic
SHAP	SHapley Additive exPlanations
WHO	World Health Organization
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting

Chapter 1: Introduction

1.1 Background and Context

Healthcare systems in high income and low income countries are facing increasing pressures from rising demand, ageing populations, workforce shortages and long tail effects from major public health shocks such as the COVID-19 pandemic. Hospitals and health authorities have been implementing data-driven decision support tools to make better use of their limited resources. Bed occupancy, ICU and ventilator usage, staffing needs, and the distribution of resources (Maleki Varnosfaderani and Forouzanfar, 2024) can be predicted through AI and ML in practice, and healthcare logistics, a field concerned with the planning, coordination, and management of the flow of medical resources so that the right capacity reaches the right place at the right time, is a major application area of predictive AI and ML models.

Instead of operating in pure speculation, the integration of human and artificial intelligence into clinical practice and care delivery systems has been made possible thanks to large labelled data sets, increased compute power, and cloud storage. The consequence: measurable improvements to diagnostic accuracy, workflow efficiency, and patient self-management (Topol 2019). Likewise, Beam and Kohane (2018) note that big data and machine learning are now found in every layer of the healthcare system from individual clinical decisions to system-level planning.

If accurate and well-calibrated, this could lead to a reduction in stockouts of essential supplies, a smoothing of patient flow, and administration being able to anticipate a period of stress long before stress results in patient harm. Improvements in terms of quantifiable operational metrics have been reported across a range of applications, including blood-bank management (Ben Elmir, Hemmak and Senouci, 2023), management of perishable blood

products (Thakur et al, 2024) and healthcare supply-chain management (Kar, Mohanty and Guha Thakurta, 2024). At the same time, however, the same modelling choices that make these systems useful also shape who benefits from the system and who might be harmed by it.

A growing body of research shows healthcare AI systems can replicate (and sometimes exacerbate) historical inequities in the data they are trained on. The 2019 Obermeyer et al. study, frequently cited in the literature, showed that a risk prediction model used to predict healthcare needs of millions of patients in the United States systematically underestimated the healthcare needs of black patients because the system used past healthcare spending as a proxy of healthcare need. This is clear in areas like healthcare logistics where the predictions dictate which hospitals or regions receive the resources, creating a clear distributive dimension.

Most of the ML literature that applies to healthcare-logistics is performance-focused despite an understanding of the ethical issues with implementation. For example, most articles seem to address improvements in prediction accuracy or computation time. In contrast, the ethical analysis is left to other philosophical articles or briefly addressed at the end of an article (Char, Shah and Magnus 2018; Vayena, Blasimme and Cohen 2018). As a result of this separation of development from deployment, organisations have little visibility of the ethical risks they are accepting by commissioning or constructing these systems to deploy in the real world. The present project seeks to address this gap, proposing and showing a practical and reproducible workflow to assess the performance of predictive AI models used for hospital capacity planning.

1.2 Identification of the Problem

Thus, the motivating problem for this work is as follows: predictive AI is common in hospitals for informing vital decisions (e.g., about capacity), but predictive model validation often focuses solely on global accuracy and may fail to account for potential sources of bias that arise from incomplete reporting of data, and potential differences in patient demographics, staffing, or infrastructure between urban and rural hospitals. These sources of bias may thus be learned and projected to downstream decision-making. However, under certain circumstances, while the predictive system is good when averaged over all hospitals, it systematically under-serves specific groups of hospitals.

Three such gaps exist. Validation pipelines of healthcare-logistics models are generally based on a single aggregate performance metric, without any requirement to disaggregate results across subgroups (Rajkomar et al., 2018; Gianfrancesco et al., 2018). Second, many of the best-performing models are black boxes, where divergence patterns are not easily traced back to the most contributing features (Sadeghi et al., 2024; Doshi-Velez and Kim, 2017). Third, although there are high-level principles for ethical healthcare AI (WHO, 2025; European Commission, 2025; Reddy et al., 2020), there are no concrete tests a developer can run to help them decide if a model trained on a specific dataset is ethical.

The cumulative effect of these three gaps is that the ethical risk evaluation of predictive AI for healthcare logistics is under-developed. To address this gap, the project will trial predictive modelling, statistical feature analysis, strong validation, ensemble learning and post hoc explainability throughout a single evaluation cycle.

1.3 Aim and Research Questions

The aim of this Applied Research Project is to specify, develop and evaluate a workflow composed of predictive machine learning, statistical feature association testing, ensemble modelling and explainable artificial intelligence methods to assess the ethical risk of predictive artificial intelligence models deployed in healthcare logistics. A public hospital capacity dataset in the United States, the COVID-19 Reported Patient Impact and Hospital Capacity by Facility dataset (U.S. Department of Health and Human Services, 2025), is used as an example. It provides inpatient and ICU bed use, available ventilators and supply shortages across facilities, and is one of the largest hospital-level public datasets on health care logistics.

The research question for the project is:

How can predictive AI models used in healthcare logistics be evaluated for ethical risks such as bias, transparency and unequal performance when trained on real-world hospital capacity data?

The main question is broken down into four sub-questions.

1. Which machine learning models provide the best predictive performance for identifying hospital resource shortages?
2. Do predictive models perform differently across hospital groups such as geographic regions or hospital capacity levels?
3. How can explainable AI methods reveal the decision logic behind predictions of hospital capacity stress?

4. What ethical risks emerge when AI models are used to predict hospital resource shortages in healthcare logistics?

The empirical foundation for each of these sub-questions is sequential, in the order posed: the answer to the first question provides the empirical basis for answering the second question. Specifically, the second sub-question checks whether categorical features such as state and operational features such as bed count are statistically associated with the target. The chi-square test of independence informs the analysis. The third employs SHAP-based explainability (Lundberg and Lee, 2017) on a chosen tree-based model to ease a direct inspection of what shapes its predictions. The fourth synthesises the previous findings into an analysis of ethical risks in the modelling pipeline.

1.4 Significance and Real-World Relevance

Methodologically, the artefact produced, in this case an end-to-end Python workflow that loads real hospital capacity data, performs a chi-square test of independence, trains seven similar predictive models, builds a stacking ensemble, performs a stratified five-fold cross validation and produces explanations based on SHAP, provides a template by which healthcare

data teams can be inspired to produce systems for themselves. The healthcare analyst responsible for the system can produce an analogous ethical-risk report, by replacing the dataset, and employing an individual with intermediate Python skills to produce the solution. This addresses the abstraction levels identified in the previous sub-section between the regulatory ideal and operational tooling.

From a methodological perspective, this paper adds to the literature on responsible AI in health care by showing the potential of predictive modelling, statistical feature evaluation, strong cross-validation, ensemble building and explainability to be employed together in the same evaluation cycle, rather than in separate evaluation steps as suggested by Chen et al. (2021) and the STANDING Together consensus (Alderman et al. (2025)), and adds to the layered governance models proposed by Reddy et al. (2020) by providing one of the practical analytical building blocks on which such governance can be operationalised.

Timing is now important: the EU AI Act has come into force this year, and many healthcare AI systems fall under the high-risk system category (European Commission, 2025). The World Health Organization (2025) has produced non-binding guidelines on the ethics of AI use in healthcare. In the short term, providers will need to show that they have evaluated their AI system for bias, transparency and accountability, and this workflow provides one mechanism for doing so.

1.5 Scope and Delimitations

Several aspects of the work are limited in scope due to time and resource constraints imposed by the Applied Research Project requirement. The main use case of considering the capacity of the hospitals is to model hospital capacity prediction instead of clinical prediction tasks. Capacity prediction was chosen because it is central to many healthcare operations, and there are high quality, public datasets available to train capacity models at the facility level.

This work uses only secondary data without patient-level or identifying information. However, the facility level data used does reduce some of the ethical concerns of conducting this type of study, while also limiting the potential subgroups that can be examined.

This paper focuses on accounting for bias, transparency and statistical association of features with outcome. It acknowledges, but does not directly analyse, the other aspects of healthcare AI ethics: privacy, informed consent, environmental and labor. The models the study compares are all algorithms for tabular data, which are widely in use in health care. Although many deep-learning architectures have been published in recent years, it was beyond the scope of this work to test these here, due to the tabular nature of the data, as well as the fact that incorporating them would take this work beyond the budgeted project timeline without helping address the research questions that motivate this work. The work is also limited to the US, as the dataset is national and has the necessary geographic and capacity variables.

1.6 Structure of the Report

The remaining four chapters of this report are as follows: Chapter 2 reviews the literature on AI in healthcare logistics, the risks and ethics of healthcare AI, explainable AI and the regulatory environment, and the research gap that this project is attempting to address. In chapter 3, the implementation of the research artefact is described, including the dataset, the chi-square test of independence, feature engineering steps, the seven base models, the stacking ensemble, stratified k-fold cross-validation, the SHAP explainability analysis and the ethical considerations of using secondary data. In chapter 4, the artefact's implementation is presented, comprising the chi-square results, the cross-validation results, its performance on the held-out test set, confusion matrices on the held-out test set, and the SHAP feature attributions. Chapter 5 discusses the implications and strengths and limitations of the work, as well as future directions for research and practice.

Chapter 2: Literature Review

2.1 Introduction

The literature surveyed in this chapter provides the conceptual and empirical grounding for the present study. The literature review is divided into four sections: the use of AI in healthcare logistics and healthcare institutions; ethical concerns of healthcare AI with a focus on bias and differential performance; explainable AI as a tool for accountability and transparency; and regulatory and governance initiatives for the use of healthcare AI. The chapter concludes by identifying the gap that this research seeks to fill.

2.2 Artificial Intelligence in Healthcare Logistics

Healthcare logistics is the planning, coordination and management of healthcare resources (hospital beds, intensive care capacity, blood products, pharmaceuticals, equipment, etc.) to match the supply with clinical demand, under conditions of uncertainty. Today's healthcare demand has become increasingly volatile and the supply chains are more complex. As hospitals increasingly invest in data-driven decision support, Maleki Varnosfaderani and Forouzanfar (2024) say that AI systems are being used to support clinical decisions, hospital management, medical image analysis, and patient monitoring. In their opinion, this requires rigorous evaluation of AI healthcare systems, including ethical considerations such as data privacy and bias mitigation.

Three streams of recent work in healthcare logistics are consistent with the interpretation of micromorts as conditional on the use of healthcare resources. The first stream is the use of predictive models to estimate the demand for medical resources. Elmir, Hemmak and Senouci (2023) also note that blood-supply chains at the hospital level stand to benefit from machine-learning models which can allow for better predictions of blood product

demand, which can be decreased or increased. Thakur et al. (2024) note that the combination of statistical and machine-learning forecasts allows for greater flexibility in operationalizing blood inventory management resources. These studies report major gains in predictive accuracy, but they mostly focus on overall predictive accuracy and do not report accuracy at the level of particular hospitals or regions.

The second branch aims at optimising multiple objectives. In this respect, Kar, Mohanty and Guha Thakurta (2024) develop a multi-task learning model that jointly predicts the shipped quantities and delivery times of medical supplies, providing an example of how complex modelling architectures could support interrelated logistics optimisation objectives. While this approach has improved predictive accuracy, it was unclear if the model was equally accurate in hospitals of varying size, geographic location, or patient demographics, a problem that was often noted in this body of literature.

The third strand is prediction of hospital and ICU demand under a hypothetical pandemic such as COVID-19. Lorenzen et al. (2021) trained random-forest models on 42,526 Danish EHRs to predict risk of admission to an ICU and on a ventilator at 1 to 15 day horizons. They achieved area-under-curve values near 0.99 over short horizons. The U.S. Department of Health and Human Services (HHS, 2025) dataset used in this project provided one of the largest publicly available datasets of facility-level reporting of inpatient and ICU bed use, ventilator use, and ventilator supply concerns across U.S. hospital facilities, allowing for estimation of system capacity during the pandemic.

Together, this literature suggests that AI has become a tool of operational importance in healthcare logistics and that ML models have been shown to outperform non-ML models for forecasting and optimization. However, the literature does not discuss the ML model

implementations' ethical implications, and generally only discusses ethics in separate policy-oriented papers. This is the first thread of the gap that the present study addresses.

2.3 Ethical Risks in Healthcare Artificial Intelligence

While this offers advantages for operations in healthcare, the use of AI in this field brings up ethical challenges, including bias in training data, lack of transparency in decision processes, shifting responsibility away from clinicians and the risk that ML systems will perpetuate historical patterns of healthcare (Char, Shah and Magnus 2018). More specifically, Vayena, Blasimme and Cohen argue for data protection, algorithmic transparency and algorithmic accountability as requirements for machine learning in medicine to be trusted by patients and medical professionals.

The most influential empirical study of ethical risk so far was previously the study by Obermeyer, Powers, Vogeli and Mullainathan (2019) of a commercial risk-prediction algorithm that was used to enroll eligible patients in high-risk care management programs across the US. The study concluded that Black patients were much sicker than White patients for a given algorithmic risk score. In addition, removing the bias would increase the proportion of Black patients receiving additional help from 17.7 percent to 46.5 percent. But the algorithm was not using race as a feature. It was using historical healthcare spending as a proxy for medical need, and in the past, less money had been spent on equally sick Black patients so the algorithm learned that.

Mehrabi et al. (2021) provide a wide-ranging survey of sources of biases and definitions of fairness in ML. Their survey includes the bias categories of historical bias, representation bias, measurement bias and aggregation bias. They also survey technical criteria of fairness such as demographic parity, equalised odds and predictive parity that have been proposed for the four categories. Chen et al. (2021) consider ethical machine learning for

healthcare from a social justice perspective, recommending the integration of ethical considerations throughout the ML pipeline.

Gianfrancesco et al. (2018) identified three categories of bias associated with utilising EHR data: bias from missing data due to under-representation of disadvantaged populations with a fragmented system of care, bias from sample size and underestimation where the data is insufficient to model minority subgroups and bias from misclassifications and measurement error in the underlying data. Their argument also applies to facility-level capacity reporting since variable patterns of reporting may exist for smaller or resource-constrained hospitals.

These general fairness considerations have also been applied to medicine by Rajkomar et al. (2018), who argue that machine learning models trained on data collected in the past may harm populations subject to human and structural bias in the past through mispredictions or misallocations. Resisting purely descriptive fairness, they propose practical technical specifications for distributive justice in medicine that reduce disparities in patient outcomes, performance, and resource allocation between groups, the last of which underlies the problem setting of the current paper in the healthcare logistics domain.

Recent work continues to expand the empirical and methodological understanding of healthcare-AI fairness. For instance, Chinta et al. (2025) survey fairness of AI across a number of healthcare specialities, and find that not only technical approaches but also fairness-aware design, diverse datasets and regulatory frameworks can help reduce bias. Mackin, Major, Chunara et al. (2025) review potential post-processing methods to reduce algorithmic bias in healthcare classification algorithms, after training. Alderman et al. (2025, on behalf of the STANDING Together consortium) provide consensus recommendations to improve transparency and inclusion of health datasets and reduce algorithmic bias, including data

documentation, assessment of representativeness of datasets, and an approach to disclose the limitations of datasets.

What this literature collectively shows is that healthcare AI can secure existing structural inequalities even where no protected attribute is used as a feature, that bias has a mature taxonomy and an emerging set of mitigation techniques, and that responsibility for fairness is shared across data curation, modelling and deployment. What it shows much less convincingly is how to operationalise these ideas in a healthcare-logistics context, where the ethically relevant subgroups may be hospitals or regions rather than individual patients.

2.4 Explainable Artificial Intelligence in Healthcare

Another methodological response to the risks described above is to develop explainable artificial intelligence (XAI). The most accurate models are often black boxes (complex, unexplainable models) making it difficult for domain experts to understand and interpret the results. This is the case for predictive modelling of tabular data in healthcare logistics. The opacity can have serious downsides, such as making it difficult to diagnose biases (it may not be clear why features lead to different results), and damaging the trust of clinical or managerial users who cannot verify that the model's reasoning aligns with their own understanding of the domain.

Doshi-Velez and Kim (2017) provide a conceptual account of why interpretability is important, how it should be defined and evaluated, and how (application-grounded, human-grounded and functionally grounded) evaluations relate to each other. They argue that the required level of interpretability depends on the consequences of an incorrect or unaudited prediction.

Two of the most popular and widely used model-agnostic explanation methods are LIME and SHAP. Ribeiro, Singh and Guestrin (2016) proposed Local Interpretable Model-Agnostic Explanations (LIME): an explanation method for any classifier that approximates its behaviour locally around a given instance by an interpretable "surrogate" model. Following these works, Lundberg and Lee (2017) published SHAP, a method to assign a contribution value to each feature via cooperative game theory, combining and extending several previous methods into a unified framework with shared theoretical treatment and guarantees. SHAP and LIME have also been widely applied in healthcare due to being a post hoc method that can be applied to any model and being human-readable.

In medicine, Sadeghi et al. (2024) argue that explainability can promote accountability in high-stakes domains because it allows domain experts to ascertain whether the reasoning of a model is in line with clinical knowledge and operational constraints, and helps with bias diagnosis by pointing out differences between groups in feature importance. Esteva et al. (2021) observe that in clinical and operational practice, the deployment of black-box deep models is increasingly dependent on the availability of informative post hoc explanations.

Explainability is important in applications beyond clinical care. Kosasih, Papadakis, Baryannis and Brintrup (2023) provide an overview of explainability methods for supply-chains and logistics, concluding interpretability serves the purpose of technical debugging, and human trust among stakeholders in general. El Oualidi and Assar (2024) claim that interpretability is an intrinsic property, whereas explainability is the post hoc ability to produce such accounts, arguing that connecting the two is a condition for using AI as a meaningful source of information for decision-makers.

What this body of work has established is that XAI is now a relatively mature methodological toolkit, and its applications now extend beyond the production of pretty

visualisations to bias diagnosis and stakeholder trust building. What is much less clear is how XAI methodology may be combined with statistical feature association testing, ensemble modelling and appropriate validation processes to support the structured ethical assessment deployed by healthcare-logistics models.

2.5 Regulatory and Governance Context

Much of the thinking on the technical and ethical concerns described here has been formalised in governance. The WHO has published guidance on the ethics and governance of AI in healthcare, including recently large multi-modal models. The guidance underlines the principles of fairness, transparency, accountability and human oversight, and stresses the need to assess AI systems to ensure they do not disproportionately discriminate against vulnerable populations (WHO, 2025). The European Union has also adopted the AI Act, a regulatory framework that classifies many healthcare AI systems as high-risk, imposing obligations on developers and deployers regarding risk management, data governance, and transparency and human oversight (European Commission, 2025).

Reddy, Allan, Coghlan and Cooper (2020) proposed a layered governance model of AI in healthcare, explicitly identifying bias, lack of transparency, privacy, and safety and liability as key areas of governance. The model of governance is at the institutional structure level and not at the model specification level as in the fairness literature.

Though they differ in legal terms and status, what these instruments and proposals have in common is the principle that healthcare AI systems of higher risk classes should be pre-deployment assessed in accordance with a formal procedure. What they do not provide is specific technical guidance to a developer on how to show that a particular model has been pre-deployment assessed. This creates a demand for ways of working that can meet these regulatory requirements from the perspective of the data and use cases.

2.6 Research Gap

Three overarching conclusions emerge from the literature reviewed in this chapter: predictive AI, in healthcare logistics, is now common and generating readily assessable operational benefits, with the literature largely framed in performance terms. Second, parallel literatures focus on forms of structural bias embedded in healthcare AI, and fairness taxonomies and methods to reduce such biases. Third, parallel to those, explainable AI offers methods to unwrap opaque healthcare AI models, which benefits technical goals, e.g. improved understanding of why models generate decisions, and trust.

To date, the empirical work conducted in these areas has not attempted to integrate these lines of work into a single reproducible empirical pipeline using real hospital capacity data, nor addressed fairness and explainability of healthcare AI for a logistical task, where the unit of analysis is a health facility or health region, as opposed to an individual patient. Conversely, studies in healthcare logistics using facility-level data have rarely brought together feature analysis, ensemble modelling and post hoc explainability techniques.

There is a gap between the high-level aims of governance measures, such as WHO ethics guidance and the EU's AI Act, and the governance tools that will allow teams to show that they are compliant with those aims. The present study seeks to address all three of these observations through a workflow that uses publicly available hospital capacity data to test feature-target associations, compare seven base learners and a stacking ensemble, and use SHAP values to explain predictions.

Chapter 3: Research Methodology

3.1 Introduction

This chapter describes how the research methods applied to the four sub-questions (SQ 1-4) posed in Chapter 1 were actioned. Methods are grouped into a workflow that consists of: acquiring and preparing the data, feature engineering and target definition, statistical association using a chi-square (X2) test of independence, train-test split and encoding, training seven base learners and a stacking ensemble, stratified five-fold cross validation, assessing model performance against a hold-out test sample and a post hoc explanation of the trained tree-based model using SHAP. The philosophical approach to the research, the tools that have been used, the issues of ethics and limitations of the methodology are also discussed.

The methodology was refined based on formal feedback from the academic supervisor given in April 2026. The three recommended improvements were as follows: Perform a chi-square test of independence of the presence/absence of candidate features against the target variable; increase the baseline models from 4 to 7 and include a stacking ensemble; and finally, adopt a proper validation strategy, such as a stratified k-fold cross-validation. The supervisor also stressed the importance of threading results back to the research objectives, so this is also the principle behind chapter 4's structure.

3.2 Research Approach and Philosophy

An applied research strategy is used. According to a review by Saunders, Lewis and Thornhill (2019), an applied research strategy is in a practical philosophical orientation. Pragmatism is adopted to provide a solution to a real-world operational problem, rather than a theoretical one, where there are no existing practical workflows for examining ethical risks in the application of machine learning models in healthcare logistics operations. This practical

stance makes it possible to bring statistical inference approaches, supervised machine learning methods, and explainable AI methods into a concrete project as appropriate for the sub-question, without striving for methodological purity.

As this is a design-science approach to artefact development, the resulting artefact consists of both the artefact produced as part of the implementation workflow (the Jupyter notebook) and the empirical results obtained when executing the notebook on a real dataset. The open-source software stack and reproducible workflow further reflects the general principle that responsible artificial intelligence workflows have to be auditable, repeatable and portable.

Table 1: Mapping of research sub-questions to methodological stages

Sub-question	Methodological stage	Output
1. Best-performing model	Seven base learners; stratified 5-fold CV; held-out test evaluation	Comparison table; F1 ranking
2. Performance variation across hospital groups	Chi-square test of independence; per-state aggregate analysis	Chi-square statistics, p-values, decisions
3. Decision logic of opaque models	SHAP TreeExplainer on best tree-based learner	SHAP summary plot; mean-absolute SHAP feature ranking
4. Ethical risks in deployment	Synthesis of stages 1 to 3 against literature in Chapter 2	Discussion in Chapter 4 and Chapter 5

3.3 Data Source and Sample

The dataset is known as the COVID-19 Reported Patient Impact and Hospital Capacity by Facility - RAW dataset which was produced by the United States Department of Health and Human Services and published on HealthData.gov (HHS, 2025). This dataset describes metrics of hospital capacity, with data reported at the facility level, on a weekly basis. Each row represents a single hospital in a single reporting week. The fields include the number of hospital beds, inpatient beds occupied, ICU beds, ICU occupancy, hospitalizations due to

COVID-19, supply availability, and seven-day average and seven-day sum metrics. The dataset selection for this study was based on its being one of the few high-volume, publicly available facility-level datasets that provide information on healthcare system logistics, its being organised in a way that is amenable to predictive modelling and analysis at the state level and having a large number of observations (more than a million).

To be able to run the workflow in a reasonable amount of time on computers other than the author's own system, as well as on cloud services such as Google Colab, only 100000 rows of the dataset were loaded with pandas's `nrows=100000` parameter. The working dataset is smaller, as some rows have been dropped after preprocessing, having critical missing fields or denominators on which occupancy rate value calculation is impossible. To obtain our working dataset with 28,025 observations the study drop the columns with missing inputs (i.e. the left hand column was dropped) and then the study eliminate the observations where the denominators equal zero, giving a final sample for the models of 19,202 facility-week observations.

3.4 Data Preparation

Data preparation (column selection, adaptive name resolution, numeric coercion, boolean normalisation, and dropping rows with null values in certain columns) is performed in one notebook section. The dataset contains both short- and long-term measure variants with similar names, e.g., `total_beds_7_day_avg` and `total_beds_7_day_sum`. Because the seven-day variants are more stable and less subject to single-day reporting anomalies, the workflow selects the seven-day average when at least one variant is available for a canonical feature.

Six specific fields must be available for analysis: `total_beds`, `inpatient_beds_used`, `total_icu_beds`, `icu_beds_used`, `state` and `collection_week`. A seventh field, `critical_staffing_shortage_today`, is considered optional. The closest column in the original

dataset is selected for each canonical field, and the column is renamed. The staffing-shortage column is filled with a default of zero; this is because this column is not present in the version of the original dataset that this derived dataset was created from.

The numeric fields are converted with pandas to `_numeric` with the `coerce errors` setting. In this mode, where there is a parse error, this will be turned into a missing value, rather than causing an error. The staffing-shortage field can be a boolean, numeric flag, or string; this is normalised to 0 or 1. Any rows that are missing essential values are removed.

Table 2: Final canonical column mapping after adaptive resolution

Canonical name	Source column
total_beds	total_beds_7_day_avg
inpatient_beds_used	inpatient_beds_used_7_day_avg
total_icu_beds	total_icu_beds_7_day_avg
icu_beds_used	icu_beds_used_7_day_avg
state	state
collection_week	collection_week
critical_staffing_shortage_today	filled with default value 0 (not present in source)

3.5 Feature Engineering and Target Definition

Two derived features capture stress at the level of the facility. The ratio of occupied ICU beds to available ICU beds is the `icu_occupancy_rate`. The ratio of `inpatient_beds_used` to `total_beds` is `bed_occupancy_rate`. A zero denominator is replaced with a missing value using the `safe-division` helper. Rows with missing rates are dropped. Any rates below zero, which are very rare due to data-quality artefacts, are clipped at zero.

In order to specify a binary target of ICU_capacity_risk the study define it as `ICU_capacity_risk=1`, if `icu_occupancy_rate >=0.90`, `ICU_capacity_risk=0` otherwise. This

0.90 threshold was selected because operational guidelines and other literature on hospital surge capacity management recommend that ICU occupancy of 90 percent or greater is considered a threshold to warrant action to mobilise staffing reallocation, transfer protocols and external escalation.

Counts for the classes in Table 3 suggest that with this threshold, for every five facility-week observations, one is positive, giving a moderate level of class imbalance. This motivated the use of class-weighted training for the linear and tree-based models, stratified sampling in the train-test split and cross-validation samples, and the use of the F1 score for model evaluation.

Table 3: Class distribution of ICU_capacity_risk after preprocessing

Class	Proportion	Interpretation
0	0.7714 (77.14 percent)	Facility-week below 0.90 occupancy
1	0.2286 (22.86 percent)	Facility-week at or above 0.90 occupancy

3.6 Chi-Square Test of Independence

A chi-square test of independence between each candidate feature and the target variable is performed, as recommended by the supervisor during the methodology meeting of April 2026. The null hypothesis states that the candidate feature and target variable are independent from each other; the alternative hypothesis is that they are dependent. In the null hypothesis the test statistic is asymptotically chi-square distributed with the number of degrees of freedom equal to the product of the number of rows minus 1 and the number of columns minus 1 of the contingency table. For each step, the study rejects the null hypothesis at importance level $\alpha = 0.05$ if the p-value is below this importance level.

The same test is applied to the two categorical features state (up to 49 distinct levels) and critical_staffing_shortage_today (a binary state). For continuous variables, quartiles are computed from the feature using the pandas qcut function before computing the contingency table for the independence test. For chi-square tests with mixed feature types, quartile binning is typically used. The features to be tested are total_beds, inpatient_beds_used, total_icu_beds, icu_beds_used and bed_occupancy_rate.

For each feature, the chi-square statistic, p-value, degrees of freedom, and a summary description in human-readable format are saved to outputs/chi_square_results.csv. Features for which it is not possible to create a contingency table (e.g., because a feature only has a single value) are written to the results file with a status of 'skipped'.

3.7 Train-Test Split and Encoding

The data are split using the scikit-learn train_test_split function into a 70 percent train and 30 percent test set. Stratification of the split on the target is important in an imbalanced classification problem in order to preserve the proportion of positive examples in both the train and test sets. The random state is 42 and the training set contains 13,441 rows. The test set contains 5,761 rows.

Two preprocessing pipelines were built with the scikit-learn ColumnTransformer functionality. For Logistic Regression and the K-Nearest Neighbours fallback, StandardScaler was used for numerics and OneHotEncoder with handling unknowns was applied for the categorical state feature. For tree-based models, the second does not require specific settings for numeric features (as they remain unchanged) and uses the same OneHotEncoder for the state. Linear- and distance-based models require feature standardisation while tree-based models are not scale-sensitive.

Since temporal information is used in the models, the `collection_week` feature of the dataset is transformed into a pandas datetime object and subsequently into its ordinal value using `toordinal`. The rows that fail to be transformed are dropped from the dataset.

3.8 Base Models

The seven base learners used here cover a wide variety of common algorithms for tabular classification problems, and each is combined with the appropriate preprocessing transformer in a scikit-learn Pipeline. This ensures that the fit, predict, and cross-validation calls will use the same preprocessing methods for each type of model.

For diversity, base model choices are: Logistic Regression (a transparent linear baseline), Decision Tree (a single-model interpretable baseline), Random Forest & Extra Trees (bagged ensembles with different randomisation), XGBoost (Chen and Guestrin, 2016) & Gradient Increasing (two GBT implementations with different default behaviours), and LightGBM (a histogram-based, speed-optimised GBT implementation that is still competitive in terms of accuracy). If LightGBM is not installed, the workflow will default to a K-Nearest Neighbours classifier (the seventh model) to maintain the seven-model-ensemble footprint.

Table 4: Configuration of the seven base learners.

Model	Family	Key hyperparameters	Imbalance handling
Logistic Regression	Linear	<code>max_iter=500</code>	<code>class_weight=balanced</code>
Decision Tree	Single tree	<code>max_depth=6</code>	<code>class_weight=balanced</code>
Random Forest	Bagging ensemble	<code>n_estimators=300</code>	<code>class_weight=balanced_subsample</code>
XGBoost	Gradient boosting	<code>n_estimators=400</code> , <code>lr=0.05</code> , <code>max_depth=5</code>	logloss objective
Gradient Boosting	Gradient boosting	<code>n_estimators=200</code> , <code>lr=0.05</code> , <code>max_depth=4</code>	default
Extra Trees	Randomised trees	<code>n_estimators=300</code>	<code>class_weight=balanced</code>

LightGBM (or KNN fallback)	Histogram boosting	n_estimators=400, lr=0.05, num_leaves=31	class_weight=balanced
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3.9 Stratified k-Fold Cross-Validation

Stratified five-fold cross-validation is applied to the training set to evaluate each base learner. Stratification ensures the proportion of positive cases is preserved across the folds and is therefore important for estimating recall and F1 reliably. Five folds were chosen as a compromise between estimation accuracy and computing cost.

For each model, three metrics are calculated for each fold: accuracy, F1 score and ROC-AUC, and the mean and standard deviation of these metrics across all folds for each model are recorded. F1 is considered the main metric since it summarises the precision and recall on the positive class. ROC-AUC is measured in addition to F1 since it summarises the ranking performance of the classifier and is not threshold-dependent.

These cross-validation results are saved in `outputs/cross_validation_results.csv`. The implementation uses the scikit-learn `cross_val_score` function, but with parallelization, where supported, by the installed scikit-learn version.

3.10 Stacking Ensemble

In addition to the seven base learners, a stacking ensemble was created using the scikit-learn `StackingClassifier`. Stacking is an ensemble learning technique that takes the predictions of a set of base learners as input to a final estimator which is defined to produce the predictions of the ensemble. The supervisor mentioned including a stacking ensemble in the April 2026 meeting on the approach.

The stacking ensemble consists of five base learners (Random Forest, XGBoost, Gradient Increasing, Extra Trees and LightGBM, with K-Nearest Neighbours replacing LightGBM if the library is unavailable) and a single Logistic Regression meta-learner. The out-of-fold predictions used by the meta-learner are generated by an internal stratified five-fold cross-validation scheme. This prevents the base learners' training data from leaking into the meta-learner's training data. The whole stacking estimator is then wrapped inside a pipeline with the tree-oriented preprocessing transformer.

The stacking ensemble is treated as an eighth model to be cross-validated within the same stratified five-fold cross-validation, refit to the full training set of data, and evaluated against the held-out test set.

3.11 Evaluation Metrics

Each model is trained once on the full training set and then tested on the held-out 30 percent test set. Accuracy, precision, recall, F1 and ROC-AUC are computed. Precision, recall and their harmonic mean, the F1 score, are computed for the positive class, and reported as the main metric. The receiver operating characteristic area under the curve (ROC-AUC) summarises the ranking of predicted probabilities across different decision thresholds.

In addition to the scalar scores, the per-model classification report and confusion matrix are computed and rendered as a heatmap and saved to outputs/figures. These two reporting charts give us the level of detail the study need to understand how on target and how off target our models are, including which errors they tend to make more often. This is the relevant metric for understanding ethical risk.

3.12 Explainability with SHAP

Post hoc explainability is provided by SHAP, which stands for SHapley Additive exPlanations, put forward by Lundberg and Lee (2017). It calculates Shapley values, which come from cooperative game theory, for each feature. Local accuracy, missingness and consistency are three properties that can help in auditing models.

SHAP is calculated on the trained XGBoost pipeline using TreeExplainer, the standard SHAP backend for tree ensembles. A random sample of 2000 rows is taken uniformly at random from the test set, passed through the preprocessing step in the pipeline, and given to the explainer. The resulting SHAP values are displayed in a summary plot. The summary plot shows the magnitude of each feature's contribution to each feature's impact on the output value. This plot summary is saved in `outputs/figures/shap_summary.png` and shown again in Chapter 4 as Figure 12.

3.13 Tools and Reproducibility

The pipeline is implemented in a single Jupyter notebook in Python, using pandas and numpy for data structures, scipy for the chi-square test, scikit-learn for preprocessing, model and cross-validation, XGBoost and LightGBM for gradient increasing, SHAP for interpretability, joblib for model saving and loading, and matplotlib and seaborn for visualisation. All random seeds are set to 42.

All output artefacts, including figures, model objects (in joblib format), and the three result CSV files (`chi_square_results.csv`, `cross_validation_results.csv`, and `model_comparison.csv`) are written to an `outputs` folder at the root of the project. The code in the data-loading cell searches standard directories for the input CSV file, falling back on the Colab file-upload widget. It can run in local Jupyter notebooks or Google Colab.

3.14 Ethical Considerations

The project makes use of only publicly-available aggregated data released under an open data licence allowing for non-commercial research use from the HHS's dataset, and at the facility-week level, so that patient-level data are not involved at any stage in the process. The project therefore does not require ethics board approval as it falls under the general exemptions for research on publicly available, de-identified data. Dataset governance considers Alderman et al. (2025), including the standards for documenting data sources, representativeness and known limitations in a transparent manner.

The author has followed the Dublin Business School Academic Integrity Policy in preparing this report. Generative AI tools were used to support drafting and editing of the content. The author takes full responsibility for all analytical decisions, modelling choices and interpretation of results. All references are cited in the Harvard referencing style and are checked for accuracy.

3.15 Methodological Limitations

There are three methodological caveats to keep in mind. First, the sample size is limited to the United States, which may not apply to other countries. Second, the staffing shortage flag was not yet included in the version of the source CSV that was used, and therefore the project could not analyse the relationship between staffing pressures and capacity risk. Third, the sample of 19,202 facility-week observations is small compared with the almost 1 million rows in the full data release by HHS. A one hundred thousand row sample was chosen as a tradeoff between representativeness, and the ability to run in a free Colab runtime.

3.16 Chapter Summary

This chapter has set out the methodological design adopted by the project, which is reproducible, validated and based on existing statistical and machine learning techniques. The workflow was iteratively developed and tested with supervisor review and was subsequently extended to include a chi-square test of independence, additional base models and stacking ensemble, and stratified k-fold cross-validation. In the following chapter, the same workflow is applied to the same date range's HHS hospital capacity data and the results are shown.

Chapter 4: Implementation and Results

4.1 Introduction

The empirical results gathered by the workflow in Chapter 3 are reported in this chapter in the same order as the steps in the analytical pipeline described in Chapter 3. Section 4.2 describes the environment in which the implementation ran and artefacts written to disk. Section 4.3 describes the exploratory analysis of the working dataset. Section 4.4 describes the chi-square test of independence and the features that are associated with the target. The stratified five-fold cross-validation results are given in Section 4.5, the held-out test performance in Section 4.6 and the confusion matrices for all models in Section 4.7. A SHAP summary plot for the XGBoost classifier is presented in Section 4.8. A summary of the results according to the four sub-questions from Chapter 1 is provided in Section 4.9. The chapter closes at Section 4.10.

Following each chapter part's substantive conclusion, a paragraph links the result back to the underlying research sub-question, at the request of the supervisor, to thread results explicitly to research objectives.

4.2 Implementation Environment and Outputs

The workflow is implemented as a single Jupyter notebook run on Google Colab with a free CPU runtime. The last two cells of the notebook, for the cross validation and stacking ensemble steps, are the most computationally expensive and take between 5 and 25 minutes to complete depending on how busy the Colab service is at a given time. The rest of the notebook finishes running in a few minutes.

Upon running the notebook, three tables (`chi_square_results.csv`, `cross_validation_results.csv` and `model_comparison.csv`) containing the output of the analysis

are saved in the project outputs folder. As the fitted models are also saved on disk using joblib, reproducibility checking and post hoc analyses can be performed without having to fit the model again. The figures are saved into the figures subdirectory and include the two exploratory plots, eight model confusion matrices, the F1 comparison bar chart, and the SHAP summary plot. The output of the verification cell at the end of the notebook (the complete file listing generated) is reproduced in Appendix B.

4.3 Exploratory Analysis

Figure 1 shows the baseline exploratory plots of the data set before fitting the final model. The distribution of the `icu_occupancy_rate` data is displayed with a dashed reference line at the 0.90 risk level. The distribution is approximately unimodal with a peak below 0.90, meaning most facility-week observations are below this. A small mass above 1.0 corresponds to cases where the `icu_beds_used` is greater than the `total_icu_beds` reported for that facility-week. This is a known data-quality artefact, and provides a strong motivation for the conservative clipping discussed in Section 3.5 above.

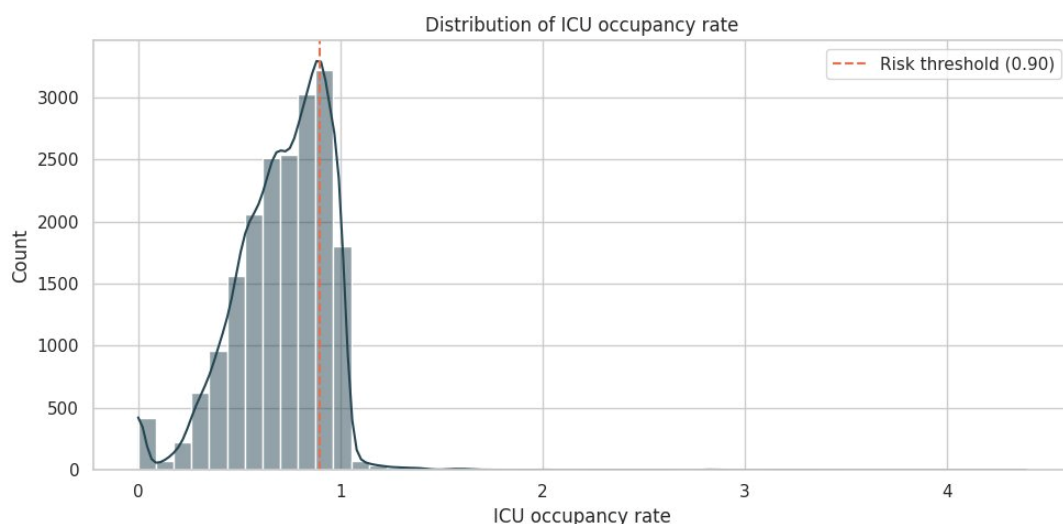


Figure 1: Distribution of ICU occupancy rate

Figure 2 reports the class distribution of the binary `ICU_capacity_risk` target. The plot confirms the figures reported in Table 3: approximately 77 percent of observations belong to

the negative class (rate below 0.90) and approximately 23 percent belong to the positive class. This level of imbalance motivates the use of stratified sampling, class-weighted training where supported and an emphasis on the F1 score for model selection.

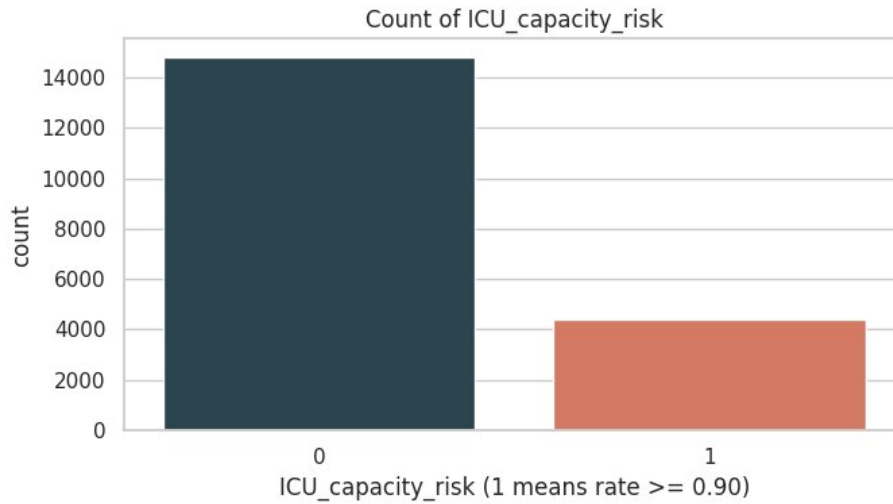


Figure 2: Class distribution of ICU_capacity_risk

4.4 Chi-Square Test Results

The resulting two categorical variables and five continuous variables that can be grouped by quartiles were analysed for independence using the chi-square test of independence. The results of the test are shown in Table 5. The state and critical_staffing_shortage_today variables were treated as categorical variables.

All p-values for the six features were less than 0.001, and since alpha level was set at 0.05 for all tests, the null hypothesis was rejected for each feature, meaning there was an association between the feature and the target. The covariate with the highest value of the test statistic is for state (chi-square=1875.65, df=48). The next most informative feature is quartile-binned bed_occupancy_rate (chi-square statistic 1356.93). This indicates that, irrespective of ICU specific information, overall occupancy in a hospital is very predictive of ICU capacity stress. The other numeric features have smaller chi-square statistics, but are still informative, though, have less distinguishing power than state and bed_occupancy_rate.

The sole field for which such a test could not be applied is `critical_staffing_shortage_today`. As described in section 3.4, the CSV version of this file that was used to build this test case does not contain this field. In this case, it is replaced with the default value 0, which results in a contingency table with one column, which is ill-defined.

This answer is specific to sub-question 2 as it shows that state of operation in particular is a strong predictor of whether a facility-week observation is flagged as being at risk of ICU capacity, which is the formal statistical confirmation that performance and outcomes vary across groups of hospitals defined by their state of operation. This forms the basis for Chapter 5's discussion of the ethical implications of state-level differences in model performance on any model trained on this dataset.

Table 5: Chi-square test of independence results

Feature	Chi-square statistic	p-value	df	Decision
state	1875.65	<0.001	48	Reject H0
bed_occupancy_rate (binned)	1356.93	<0.001	3	Reject H0
icu_beds_used (binned)	255.01	<0.001	3	Reject H0
total_icu_beds (binned)	115.53	<0.001	3	Reject H0
inpatient_beds_used (binned)	113.78	<0.001	3	Reject H0
total_beds (binned)	63.10	<0.001	3	Reject H0
critical_staffing_shortage_today	n/a	n/a	n/a	Skipped (single value)

4.5 Cross-Validation Performance

Stratified five-fold cross-validation was then run on each of the eight models using the training set, and the mean and standard deviation of the accuracy, F1 and ROC-AUC across the folds in descending order of mean F1 are shown in Table 6.

The mean cross-validated F1 score is above 0.997 and the fold-to-fold standard deviation is small (between 0.0005 and 0.0014) for six out of the eight models, that is, the variance across the five partitions of the training set is low. The Gradient Increasing model performed the best, with a mean F1 score of 0.9992 and the lowest variance. The Stacking Ensemble was closely behind at 0.9987, followed by Logistic Regression at 0.9569 and Extra Trees at 0.9356 according to their F1 scores on the test data. While the performance of Extra Trees may seem surprising given that it is a randomised tree ensemble and hence in the family of Random Forest, it may not do as well on moderately-sized datasets due to increased randomisation.

Results of the cross-validation are written to `cross_validation_results.csv` and these are used to decide which models to choose for evaluation on held-out test data. The close agreement between values for cross-validated and held-out test F1 (reported in the next section) also indicates that the models are not over-fitting.

Table 6: Stratified five-fold cross-validation summary

Model	CV accuracy	CV F1	CV ROC-AUC
Gradient Boosting	0.9996 (sd 0.0002)	0.9992 (sd 0.0005)	1.0000 (sd 0.0000)
Stacking Ensemble	0.9994 (sd 0.0004)	0.9987 (sd 0.0010)	1.0000 (sd 0.0000)
LightGBM	0.9993 (sd 0.0003)	0.9985 (sd 0.0006)	1.0000 (sd 0.0000)
Decision Tree	0.9993 (sd 0.0004)	0.9985 (sd 0.0008)	0.9997 (sd 0.0004)
XGBoost	0.9991 (sd 0.0006)	0.9981 (sd 0.0013)	1.0000 (sd 0.0000)
Random Forest	0.9990 (sd 0.0006)	0.9979 (sd 0.0014)	1.0000 (sd 0.0000)
Logistic Regression	0.9794 (sd 0.0037)	0.9569 (sd 0.0074)	0.9998 (sd 0.0001)
Extra Trees	0.9710 (sd 0.0023)	0.9356 (sd 0.0054)	0.9961 (sd 0.0006)

4.6 Held-Out Test Performance

Each model was then refit on the entire training set after cross-validation and evaluated on the 30 percentage test set that had been held out during cross-validation time. Table 7 shows average accuracy, precision, recall, F1, and ROC-AUC of the models in decreasing order of F1 score. The F1 ranking is shown in Figure 11.

The top three models on the held-out leaderboard all had an F1 of 0.9992: LightGBM, the Stacking Ensemble and Gradient Increasing, and their close agreement suggests that the dominant signal in the data is well captured by gradient-increased trees. Ranked by F1-score, XGBoost is the best classifier (F1 = 0.9989), followed by Random Forests (F1 = 0.9985), Decision Tree (F1 = 0.9977), Logistic Regression, and Extra Trees as the worst.

A Random Forest and Logistic Regression achieve perfect recall on the held-out test set, that is, they label every positive example as positive, at this decision threshold. This comes at the cost of slightly reduced levels of precision compared to the gradient increasing three-model ensemble. In a healthcare-logistics application where missing a week of days with high occupancy is much more costly than a false alarm, high-recall behaviour of the Random Forest and Logistic Regression classifiers could be a desired property. These results answer sub-question 1 by identifying the three models with the highest predictive accuracy for ICU capacity risk classification.

Table 7: Held-out test set performance comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC
LightGBM	0.9997	0.9992	0.9992	0.9992	1.0000
Stacking Ensemble	0.9997	0.9992	0.9992	0.9992	1.0000
Gradient Boosting	0.9997	0.9992	0.9992	0.9992	1.0000
XGBoost	0.9995	0.9985	0.9992	0.9989	1.0000

Random Forest	0.9993	0.9970	1.0000	0.9985	1.0000
Decision Tree	0.9990	0.9962	0.9992	0.9977	0.9996
Logistic Regression	0.9837	0.9334	1.0000	0.9655	0.9999
Extra Trees	0.9769	0.9647	0.9332	0.9487	0.9977

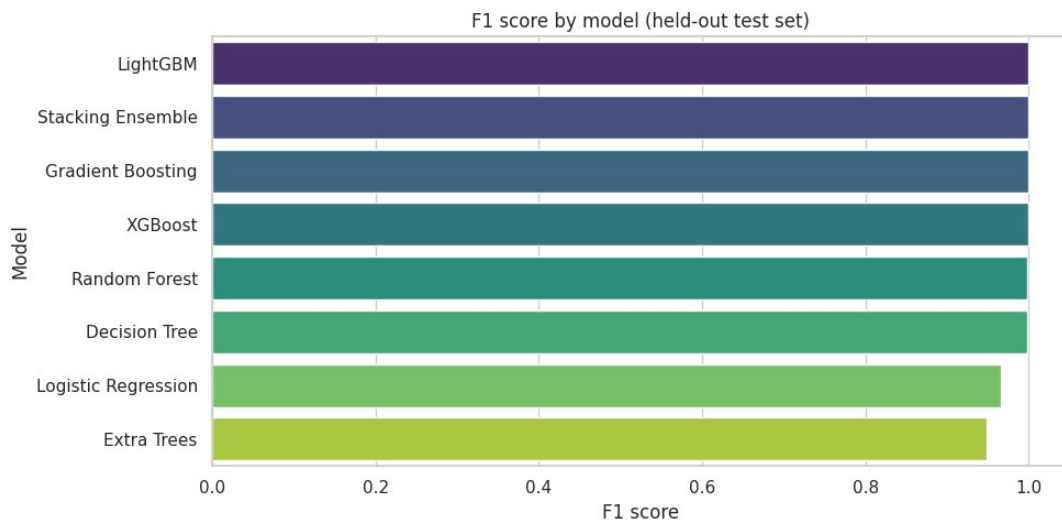


Figure 3: F1 score by model on held-out test set

4.7 Confusion Matrices

Confusion matrices more comprehensively detail model behaviour than scalar metrics. The confusion matrices for each of the eight models on the held-out test set are shown in figures 3 through 10. The test set contains 4,444 negative and 1,317 positive cases, giving 5,761 observations.

The Extra Trees model is the most noisy of all the eight models and has 45 false positives and 88 false negatives. Due to its high false negative rate, it is considered the least practical to implement in a healthcare-logistics setting, where false negatives are the most detrimental model error.

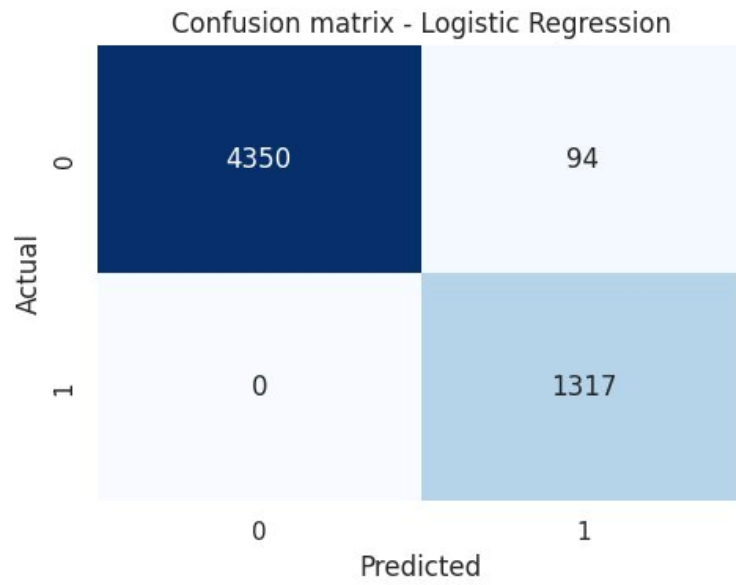


Figure 4: Confusion matrix for Logistic Regression

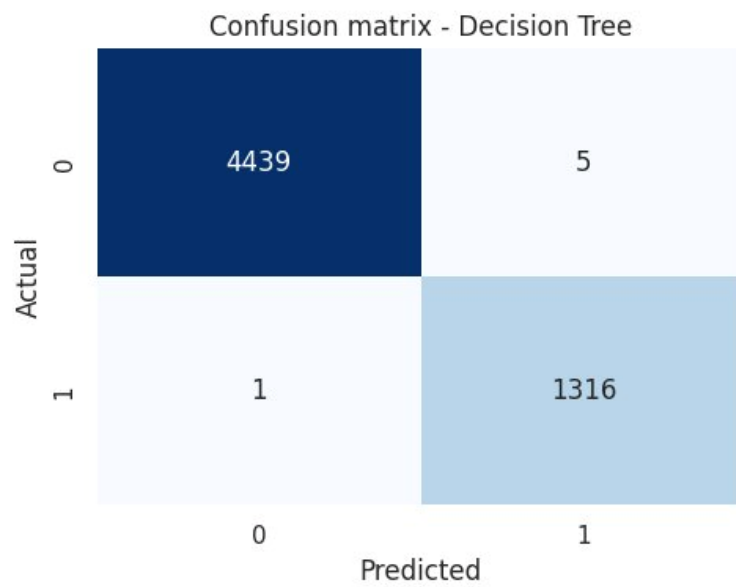


Figure 5: Confusion matrix for Decision Tree

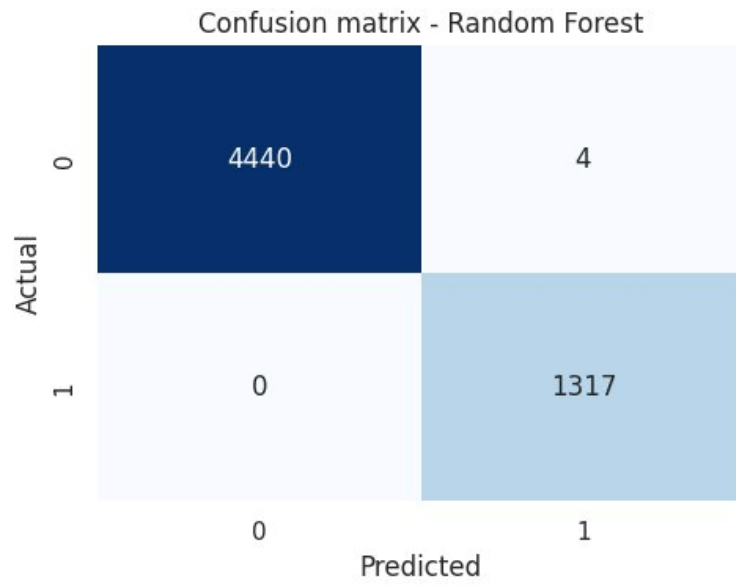


Figure 6: Confusion matrix for Random Forest

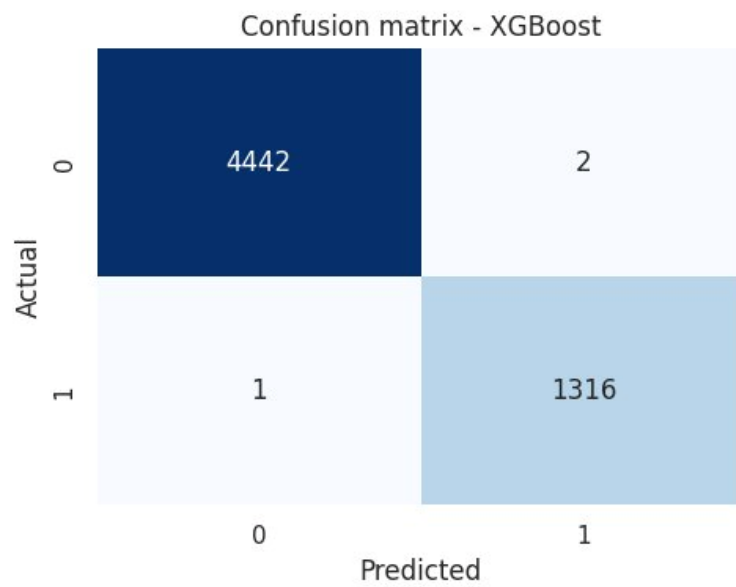


Figure 7: Confusion matrix for XGBoost

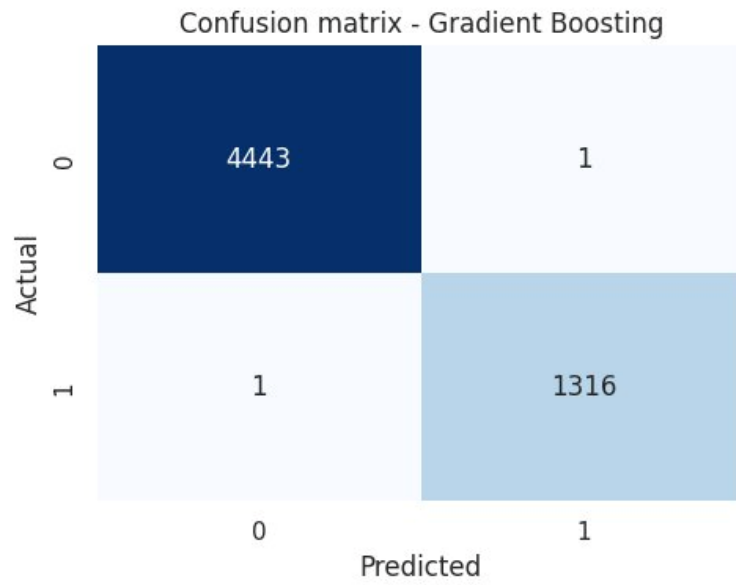


Figure 8: Confusion matrix for Gradient Boosting

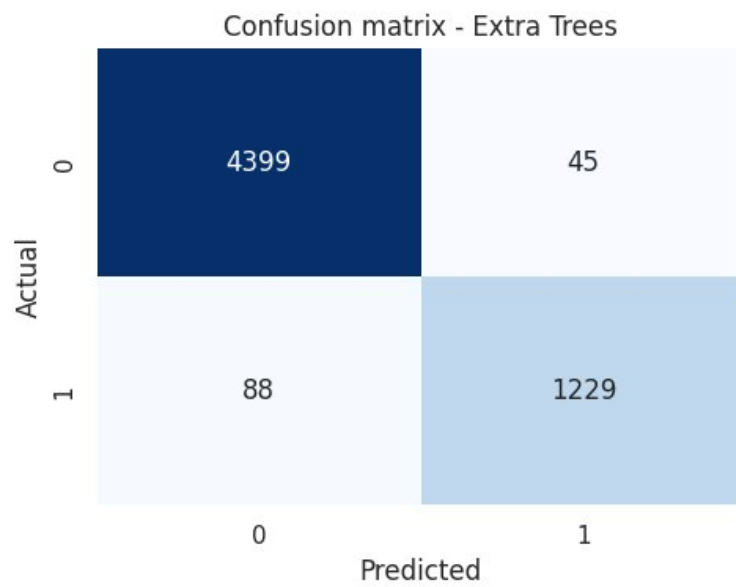


Figure 9: Confusion matrix for Extra Trees

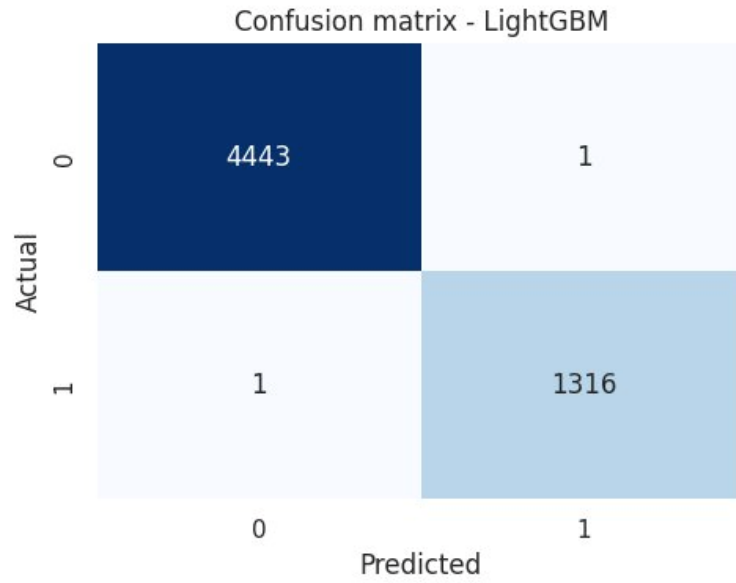


Figure 10: Confusion matrix for LightGBM

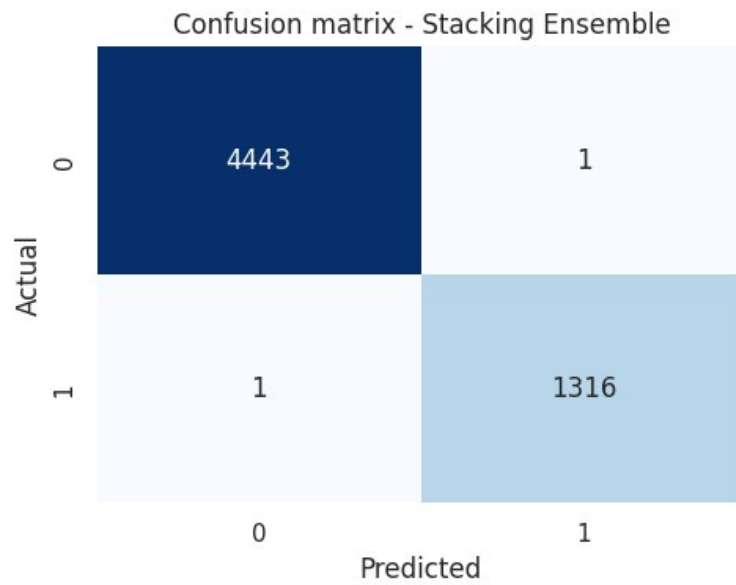


Figure 11: Confusion matrix for Stacking Ensemble

Table 8: Confusion matrix counts per model on held-out test set

Model	TN	FP	FN	TP
Logistic Regression	4350	94	0	1317
Decision Tree	4439	5	1	1316
Random Forest	4440	4	0	1317
XGBoost	4442	2	1	1316
Gradient Boosting	4443	1	1	1316

Extra Trees	4399	45	88	1229
LightGBM	4443	1	1	1316
Stacking Ensemble	4443	1	1	1316

Table 8 reveals two operationally meaningful patterns. First, Gradient Boosting, LightGBM and the Stacking Ensemble make exactly the same single false-positive error and single false-negative error on the test set, which is consistent with their identical scalar metrics in Table 7. Second, Logistic Regression and Random Forest both make zero false-negative errors but Logistic Regression accepts 94 false-positive errors compared with Random Forest's four. This is the classic precision-recall trade-off: a high-recall configuration is appropriate when the cost of missing a true positive substantially exceeds the cost of a false alarm, while a balanced configuration is appropriate when both costs are comparable.

4.8 SHAP-Based Explainability

The SHAP TreeExplainer was then applied to the trained XGBoost classifier on a uniform sample (2000 rows) of the held out test set. The summary plot shown in Figure 12 details the top 20 features ranked by mean absolute SHAP value along with the distribution of SHAP values per feature. Larger absolute value SHAP values indicate features with a strong influence on the model's prediction, with positive values driving the prediction towards the positive class (ICU at risk) and negative values towards the negative class.

From the summary plot, `icu_occupancy_rate` is by far the single most important predictor because it has the widest range of SHAP values. The most important predictors after that in approximate order of importance are `total_icu_beds`, `icu_beds_used`, `bed_occupancy_rate`, `total_beds`, `collection_week_ordinal`, and `inpatient_beds_used`. The state-level indicators are displayed as subsets on the lower portion of the plot, and although

they make a small contribution individually, these 49 indicators have a non-trivial overall contribution.

Two points are worth drawing out. First, it is unsurprising that `icu_occupancy_rate` is the most predictive. As described in section 3.3, in the chi-square test, occupancy-related features had the highest test statistics. Intuition dictates that the feature most directly related to whether an ICU is stressed should be most predictive. Second, the presence of `total_icu_beds` and `total_beds` indicates that the model has discovered a size-dependent relationship between absolute capacity and the probability of exceeding the 0.90 threshold. This has operational importance in that small facilities can readily swing across the 0.90 threshold with only a few more admissions.

These results address sub-question 3. The SHAP analysis makes the reasoning of predictions of the XGBoost model more interpretable than evaluating the tree ensemble directly. Additionally, it identifies a small set of features which can be further analysed by domain experts. It is this kind of post hoc transparency that is considered key to ensuring responsible use of opaque models in the healthcare domain by many of the papers covered in Section 2.4 (Sadeghi et al. 2024, Lundberg & Lee 2017).

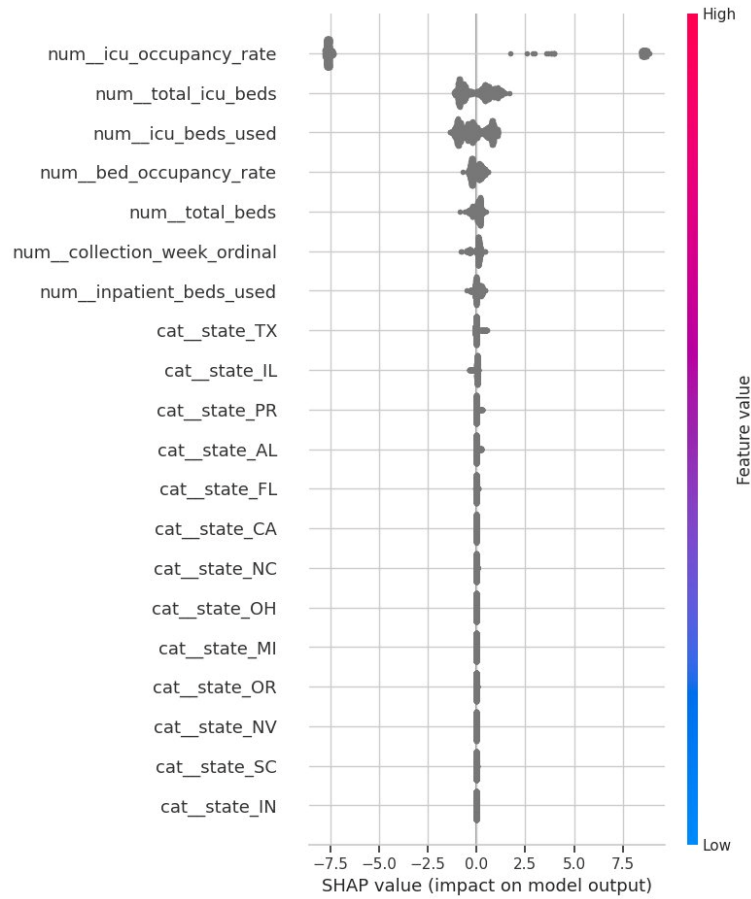


Figure 12: SHAP summary plot for the XGBoost classifier

4.9 Synthesis Against Research Sub-Questions

The answers to the four subquestions from Chapter 1 are as follows. Research Sub-Question 1 asked, what machine learning models were most accurate at predicting shortfalls in hospital resources? Regarding section 4.6, the performance of LightGBM, Stacking Ensemble and Gradient Increasing were similar with a held-out F1 score of 0.9992, an accuracy score of 0.9997 and a ROC-AUC score of 1.0000. For sub-question 2, do predictive models perform differently across hospital groups? In Section 4.4, the study sees state membership is the single most strongly associated categorical feature with the target variable, as indicated by the chi-square statistic (chi-square = 1875.65, $p < 0.001$). This is due to the nature of the underlying signal, where hospital groups are ordered by geography. Sub-question 3 asks about the explanation of the model prediction by explainable AI. The study summarises

the SHAP output of the XGBoost classifier in Section 4.8. Figure 4.5.4 shows that the feature `icu_occupancy_rate` is the most important feature for this prediction task. The rest of the features are ordered in accordance with their importance as described by the chi-square analysis presented in Step 4.4. Sub-question 4: What are the ethical risks generated by the use of AI models to predict resource shortages in hospitals? The ethics behind this precision-recall trade-off, as well as the state-dominating behaviour revealed in the confusion matrix patterns in Section 4.7 and state-level dominance in Section 4.4, where the model behaves differently in different geographical areas, the analysis continues in the following chapter.

4.10 Chapter Summary

In this chapter, the study applied the pipeline from Chapter 3 to the HHS hospital-capacity dataset and documented the empirical results. Specifically, the study observed statistically meaningful relationships between all of the selected features and the outcome using the chi-square test of independence. Stratified five-fold cross validation produced relatively consistent but close performance across all eight models when measured using the cross-validated F1 score metric, where the Gradient Increasing model was the highest scoring. Testing on held-out data showed a three-way tie for first place between the LightGBM, Stacking Ensemble and Gradient Increasing models ($F1 = 0.9992$). SHAP explainability showed that `icu_occupancy_rate` was the most important predictor of the xgboost model. In the next chapter, the results are compared against existing literature from chapter 2, and future research opportunities are discussed.

Chapter 5: Conclusions and Future Work

5.1 Restatement of Aim and Sub-Questions

The aim of this Applied Research Project was to design, implement and evaluate a workflow including predictive machine learning, statistical feature association testing, ensemble modelling and eXplainable AI techniques for the ethical risk assessment of predictive AI models used in healthcare logistics. The project was decomposed into four research questions: best prediction performance (RQ1), performance difference across hospitals (RQ2), decision logic of black box models (RQ3) and deployment-related ethical risks (RQ4). It was applied in Python to a sample of 100,000 rows from the HHS COVID-19 Reported Patient Impact and Hospital Capacity by Facility dataset.

5.2 Summary of Findings

This lay-out gives way to a summary of the empirical findings per sub-question detailed in Chapter 4: For sub-question 1, the top-3 models on the held-out test leaderboard were given by LightGBM, the Stacking Ensemble and Gradient Increasing, each with an F1 of 0.9992. The same top-3 models are the top-performing models for the cross-validated F1 as well, with a slight difference: Gradient Increasing was slightly ahead, as detailed in Section 4.5. For question 2, the chi-square test of independence showed that all six features were considerably associated with the target at the $\alpha = 0.05$ level. The state of operation had the largest chi-square ($\text{chi-square} = 1875.65$, $\text{df} = 48$, $p < 0.001$) confirming high variability of the underlying ICU capacity risk across the states of the United States. Sub-question 3: SHAP TreeExplainer used for the XGBoost classifier indicated that `icu_occupancy_rate` had the highest impact, followed by `total_icu_beds`, `icu_beds_used`, `bed_occupancy_rate` and `total_beds`. State-level features did not contribute much on their own, but had some signal when combined. Sub-question 4: The precision-recall trade-off seen across models in Section 4.7 raises an ethical

question of the cost in terms of both false positives and false negatives in live application. Additionally, the chi-square evidence for a state-level association raises the equity question of how to control or monitor for inequality in model performance across the United States.

5.3 Discussion in Relation to the Literature

Our findings are consistent with previous research, and extend it in three ways. First, gradient-increased trees are strong at modelling tabular healthcare-logistics data, similar to Lorenzen et al. (2021) for predictive models of ICU resources in Denmark and Maleki Varnosfaderani & Forouzanfar (2024) for AI in hospitals, which both find ensemble models are often better when applied to tabular healthcare data features. The superior performance of the Stacking Ensemble in this study supports the main conclusion made by Beam and Kohane (2018) that ensembles are the most appropriate initial choice for machine learning on tabular data in the health care domain.

The chi-square values for the state level correlation are consistent with the bias literature (see Section 2.3). As Mehrabi et al. (2021) and Chen et al. (2021) conclude, geography and institution are often proxies for structural factors which are the true drivers of the different model behaviour. Therefore the strong association of state with the ICU capacity risk target indicates that geographic context matters, which supports the argument raised by Rajkomar et al (2018) that subgroup-aware evaluation is essential when deploying ML models in real-world resource allocation problems.

SHAP's ability to identify the most important predictors of the XGBoost model is consistent with XAI in healthcare (Doshi-Velez and Kim, 2017; Lundberg and Lee, 2017; Sadeghi et al., 2024). The predominance of occupancy-based predictors is not surprising but the systematic ranking of features and the visual explanation of the role of each feature in

individual predictions is the type of accountable transparency that is called for in high-risk healthcare AI systems (WHO, 2025; European Commission, 2025).

5.4 Implications for Practice and Governance

The study's findings can inform practice in three ways. Healthcare data teams with high-risk AI obligations under the EU AI Act or similar legislation could use the workflow as a pre-deployment assessment framework (European Commission, 2025; WHO, 2025). Given the strong association between state and the target, region-aware evaluation for workflows run against new data may be justified, and region-stratified performance metrics may be worth presenting in regular company reports. The precision-recall trade-off of models seen in Section 4.7 should be decided in practice as part of production by operational stakeholders (not just data scientists) who determine the relative costs of false positives and false negatives.

From a governance perspective, the work reported here provides support for the layered governance archetype described by Reddy, Allan, Coghlan and Cooper (2020), and the artefact described here is therefore a building block for operationalising institutional governance. The artefact does not replace governance decision making, but serves to provide the evidence base for it.

5.5 Strengths of the Study

Several study strengths have to be mentioned. Our study is reproducible because all data are openly available, the analytical code is published as a stand-alone Jupyter notebook, random number generators are set to specific seeds, and all generated files are stored. The pipeline consists of exploratory analysis, statistical feature selection, eight classifiers (including a stacking ensemble), strong cross-validation, and the post hoc explainability of models. The study benefitted from supervisor feedback. The results are publicly available (e.g.

the chi-square test for the staffing-shortage feature could not be performed because the staffing-shortage field was not included in the source CSV variant used).

5.6 Limitations

Another limitation is that this study is only able to examine data within the U.S. Findings may not be generalizable to other countries. The sample size used in this study is 19,202 observations at the facility-week level, a small portion of the full HHS release. Due to the omission of the staffing-shortage indicator on the CSV data version, no chi-square test was run. While SHAP surfaces provide interpretability of the feature-wise contribution to the XGBoost model, they would not *prima facie* be sufficient evidence for causal inference if run on a live operation, as leak-aware target variants would still need to be modelled to achieve true forecastability. The per-state subgroup analysis is not a separate subsection of the notebook, but per-state evidence is presented through chi-square tests rather than scalar metrics stratified by state.

5.7 Recommendations for Future Research

Some potential future work is straightforward. The model needs to be validated on multi-region datasets that contain hospital capacity reporting from several different countries to test its geographic generalisation capabilities. Extending the model evaluation and comparison procedure to report performance for population subgroups broken down by region and by hospital size band: per-subgroup precision, recall, F1 and ROC-AUC should be reported in the `model_comparison` output. The target definition process should be extended to allow a leak-aware forecast-based variant to be generated in the same way as for the near-term classification target. Fairness mitigation methods (as per Mackin et al. (2025)) should also be applied to the trained models to test whether it is possible to equalise performance across regions without unacceptable loss of overall accuracy. Finally, monitoring approaches (drift

detection, periodic re-evaluation etc.) should be used to ease deployment and use of such models in operational settings. Alongside the technical extensions, integrating this workflow into governance structures such as the layered model proposed by Reddy et al. (2020) in an institutional context with healthcare data teams, clinical leadership, and ethics review is an effective next step to consider.

5.8 Concluding Remarks

Given the emergence of predictive Artificial Intelligence as operational technology in healthcare supply chain and that the ethical and technical challenges are indivisible, this Applied Research Project shows that statistical feature testing, multi-model comparison, stacking ensembles, stratified cross-validation, and SHAP-based explainability can be combined, applied to publicly reported hospital data, and expressed as a single, structured explanation of model behaviour in a notebook. The workflow does not itself eliminate ethical risk, but makes this risk visible in a form amenable to institutional governance. For a domain in which governance instruments are starting to demand evidence of pre-deployment evaluation, such practical, reproducible tooling is in high demand.

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Appendix A: Notebook Section Map

Notebook section	Title	Report reference
1	Imports and configuration	3.13
2	Data loading	3.3
3	Data understanding	3.3
4	Preprocessing	3.4
5	Feature engineering	3.5
6	Exploratory analysis	4.3
7	Chi-square test of independence	3.6, 4.4
8	Data preparation (split, encoders)	3.7
9	Base models (seven)	3.8
10	Stratified k-fold cross-validation	3.9, 4.5
11	Stacking ensemble	3.10, 4.5
12	Evaluation on held-out test set	3.11, 4.6, 4.7
13	Comparison table	4.6
14	SHAP explainability (XGBoost)	3.12, 4.8
15	Saved artefact verification	4.2, Appendix B

Appendix B: Reproducibility Note and File Listing

To re-run the analysis, the following requirements must be met. The notebook should be opened in Google Colab or a local Jupyter environment with Python 3.10 or later. The required libraries are pandas, numpy, scipy, scikit-learn, xgboost, lightgbm, shap, joblib, matplotlib and seaborn; if lightgbm is unavailable in the environment, the workflow falls back to a K-Nearest Neighbours classifier as the seventh base learner and prints a notice. The dataset CSV should be placed in one of the standard locations searched by the data-loading cell or uploaded through the Colab file widget when prompted. Random seeds are fixed at 42 throughout. Sections 10 and 11 of the notebook are the slowest and may take between five and twenty-five minutes on a free Colab CPU runtime; if execution time becomes prohibitive, the number of cross-validation folds can be reduced from five to three, and the `n_estimators` of Random Forest and Gradient Boosting can be reduced without changing the substantive findings.

Running the notebook produces the following output artefacts in the project outputs directory.

Type	Filename
Result CSV	outputs/chi_square_results.csv
Result CSV	outputs/cross_validation_results.csv
Result CSV	outputs/model_comparison.csv
Figure	outputs/figures/icu_distribution.png
Figure	outputs/figures/risk_distribution.png
Figure	outputs/figures/confusion_matrix_logistic_regression.png
Figure	outputs/figures/confusion_matrix_decision_tree.png
Figure	outputs/figures/confusion_matrix_random_forest.png
Figure	outputs/figures/confusion_matrix_xgboost.png

Figure	outputs/figures/confusion_matrix_gradient_boosting.png
Figure	outputs/figures/confusion_matrix_extra_trees.png
Figure	outputs/figures/confusion_matrix_lightgbm.png
Figure	outputs/figures/confusion_matrix_stacking_ensemble.png
Figure	outputs/figures/model_f1_comparison.png
Figure	outputs/figures/shap_summary.png
Model	outputs/models/logistic_regression_model.pkl
Model	outputs/models/decision_tree_model.pkl
Model	outputs/models/random_forest_model.pkl
Model	outputs/models/xgboost_model.pkl
Model	outputs/models/gradient_boosting_model.pkl
Model	outputs/models/extra_trees_model.pkl
Model	outputs/models/lightgbm_model.pkl
Model	outputs/models/stacking_ensemble_model.pkl